

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 150

Hum 4441: Engineering Ethics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

- ✓1. The second article examines how intellectual property (IP) protection in strengthens enterprise-led Industry-University-Research Institution (E-IUR) cooperation within China's National Innovation System (NIS). It focuses on how enterprises work as key drivers of innovation and examines how IP protection policies, particularly the Intellectual Property Demonstration Cities (IPDC) policy, influence collaboration outcomes. Using data from Chinese listed companies (2008–2020) and a difference-in-differences (DID) approach, the study evaluates IPDC's effect on E-IUR integration, measured via prestigious innovation awards such as the State Science and Technology Progress (SSTP) Award. Findings show that the IPDC significantly boosts E-IUR cooperation by increasing enterprise R&D investment and strengthening regional IP law enforcement, reducing risks of imitation and knowledge spillover. The study also highlights the asymmetric power dynamics within IUR cooperation, where enterprises often dominate due to resource dependence, challenging assumptions of equal collaboration among innovation actors. Unlike prior studies that focus mainly on patents or R&D spending, this research uses award-winning innovations as a proxy, capturing not just invention but also commercialization potential. Results confirm that IP protection policies can enable deeper collaboration, technological progress, and industrial competitiveness. The study recommends that governments should expand and refine IPDC initiatives, establish best practices for IP protection, and encourage broader participation by cities. Ultimately, robust IP systems are needed for national and regional innovation capacities, particularly in emerging economies striving to **catch up** technologically. Now, answer the following:
- a) How does the IPDC policy strengthen the deep integration of E-IUR cooperation in China? 12.5 (CO1) (PO6)
- b) Why does the study use innovation awards, rather than patents or R&D spending, as a measure of E-IUR cooperation effectiveness? 12.5 (CO1) (PO6)
- ✓2. Huawei, a leading technology company, drives 5G network development and supplies smartphones and digital infrastructure globally but faces scrutiny over data privacy and ethics. Concerns from governments and organizations suggest Huawei's technology could enable unauthorized data collection or surveillance, though the company denies these claims. From an ethical standpoint, the situation raises questions about corporate responsibility, transparency, and trust. Companies like Huawei are not only expected to innovate and provide secure, efficient technologies, but also must also ensure strong safeguards to protect the user privacy, respect international data laws, and maintain public trust. This case provides a real-world example of the ethical dilemmas faced by global tech companies: balancing innovation, market expansion, and profitability with the moral obligation to protect the user rights and data privacy. Now, answer the following:
- a) If you were a policymaker, what ethical responsibilities would you expect Huawei (or any global tech company) to uphold in handling user data and privacy? 12.5 (CO4) (PO6)
- b) As an engineer working at Huawei, how should you respond if you discover practices that may compromise user privacy for business or political interests? 12.5 (CO1) (PO8)

3. Amazon Simple Storage Service (Amazon S3), supports millions of businesses by storing sensitive data like financial records and customer details. Despite robust security features like encryption and access controls, misconfigured S3 buckets have led to data breaches, exposing private information due to companies errors in setting access permissions, not flaws in Amazons system. Still, the responsibility question arises: who is accountable for protecting user data, the service provider (Amazon) or the client company using the service? From an ethical perspective, engineers working with cloud systems must balance efficiency, cost, and innovation with the duty to safeguard data privacy. Mistakes in configuration may appear technical, but have serious ethical implications, putting users at risk of fraud, identity theft, or loss of trust. The Amazon S3 case highlights the critical role of ethical responsibility, professional diligence, and transparency in cloud-based data management. Now, answer the following:

a) In cases of S3 data breaches via misconfiguration, should the ethical responsibility lie more with Amazon (as the provider) or with the client company (as the user)? Justify your answer.

12.5
(CO4)
(PO6)

b) As a future computer engineer, how would you ensure that cloud-based services like S3 are used ethically and securely in real-world projects?

12.5
(CO1)
(PO3)

4. In computer science and engineering, many innovations are not only technical but also social in nature, affecting millions of users. For example, social media platforms, AI-driven recommendation systems, and large-scale data analytics projects are often released in society as “experiments.” Companies sometimes test new algorithms or features directly on users without their explicit consent, treating society as a large-scale “lab.” One notable example occurred when a social media company altered the user news feeds to study emotional responses, without informing them. Similarly, ride-sharing and e-commerce platforms have run experiments to see how pricing algorithms, wait times, or product recommendations influence user behavior. While these studies can generate valuable insights, they also raise ethical dilemmas:

- i. **Consent:** Is it ethical to experiment on users without informing them?
- ii. **Risk:** Could these experiments cause harm (e.g., stress, manipulation, discrimination)?
- iii. **Responsibility:** Who is accountable when social experimentation causes unintended consequences — engineers, managers, or the organization?

For computer science engineers, this highlights the tension between innovation and responsibility. Engineers must consider not only technical feasibility but also the broader societal impact of deploying experimental technologies. Social experimentation thus becomes a matter of engineering ethics, where issues of transparency, accountability, and respect for human rights are central. Now, answer the following:

a) Should engineers be ethically obligated to get user consent before deploying large-scale social experiments (e.g., algorithm changes, A/B testing)? Why or why not?

12.5
(CO2)
(PO6)

b) How can computer engineers balance innovation (pushing forward new technologies) with social responsibility when experimenting with systems that directly affect human behavior?

12.5
(CO3)
(PO6)

5. A growing debate in computer science and engineering is whether technology should primarily serve social good or economic profit. Many companies present their innovations as solutions to global challenges, but in practice, commercial interests often dominate. Consider a popular social media platform. Its official mission is to “connect people and communities,” which reflects a tech for good vision. However, its main source of revenue comes from targeted advertising, driven by algorithms that prioritize engagement. These algorithms sometimes promote sensational or polarizing content, raising concerns about misinformation, mental health, and social

division. A similar tension exists in the healthcare technology sector. For example, AI tools can diagnose diseases faster than doctors, potentially saving lives (tech for good). Yet, some companies prioritize monetizing patient data or restricting access to their AI systems to wealthier markets (tech for profit), limiting benefits for vulnerable populations. This creates an ethical dilemma for engineers: Should they design systems that maximize profit for their company, or should they advocate for solutions that prioritize long-term social well-being? While profit ensures financial sustainability and further innovation, prioritizing only profit can harm users, deepen inequality, and undermine trust in technology. For computer engineers, the challenge is not choosing one side completely, but finding a responsible balance between innovation for profit and innovation for social good. Based on this case, answer the following questions:

- a) How should engineers and technology companies balance the ethical responsibility of developing "tech for good" with the economic pressures of generating "tech for profit"? 12.5 (CO1) (PO8)
 - b) In cases where profit-driven decisions may harm society (e.g., spreading misinformation, limiting healthcare AI access), what ethical responsibilities do engineers have to intervene? 12.5 (CO1) (PO6)
6. Digital forensics involves investigating digital devices to uncover evidence of crimes like hacking, fraud, cyberbullying, or data theft, playing a crucial role in law enforcement and cybersecurity. However, it raises significant ethical issues. For instance, consider a university's internal investigation into a data breach, where a forensic team is tasked with examining students' laptops and cloud accounts for suspected hacking into the grading system. The investigation uncovers evidence of unauthorized access, but it also reveals large amounts of unrelated personal data: private emails, photos, and medical records of students. This creates an ethical dilemma:
- i. **Scope of investigation:** Should forensic analysts strictly limit their search to relevant data, or can they use unrelated personal information if it reveals misconduct?
 - ii. **Privacy vs. security:** Protecting the university system is important, but should it come at the cost of violating individual privacy?
 - iii. **Bias and fairness:** If investigators misuse or over-interpret evidence, it could unfairly damage the career of a student.

In real-world cases, digital forensic experts often face similar dilemmas. Their work **must** balance justice and truth-seeking with respect for privacy, fairness, and professional integrity. For CSE engineers, this highlights the ethical responsibility of using powerful forensic tools responsibly, transparently, and with clear legal and ethical boundaries. Based on this case, answer the following questions:

- a) In digital forensic investigations, how should engineers and investigators ethically balance the need for security and justice with the protection of individual privacy rights? 12.5 (CO4) (PO8)
- b) What ethical responsibilities do digital forensic professionals have if they discover sensitive but unrelated personal information during an investigation? 12.5 (CO4) (PO8)